Amrita School of Computing
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[Editor-in-Chief]
*[journal name]*

Dear [Editor-in-Chief],

We submit "Chunking Without Leaking: Provably Equivalent, Bounded-Memory Redaction of Personal Data and Credentials" for consideration as an original research article in *[journal name]*.

Logs, database exports and packet captures must be stripped of personal data and credentials before they are shared, and they are routinely too large to process in one pass. Chunked processing introduces a silent leak: a secret that straddles a chunk boundary is missed and written out in the clear. We found no redaction, de-identification or secret-scanning system that states when its chunked output equals its whole-document output.

The paper closes that gap for context-dependent redaction. Each detector declares the longest span it emits and the context it reads; we show that an overlap as long as the longest span can still leak, and prove that an overlap covering every detector's span plus its right context makes streamed output equal to whole-document output, that the parallel engine is byte-identical to the sequential one or falls back to it, and that working memory is independent of input size. All proofs are in the paper (Section 5 and Appendix A); Online Resource 1 maps each step to the implementation and holds supplementary tables. The default overlap meets the requirement exactly. Measured memory stays between 8.4 and 8.54 MiB from 8 MiB to 2 GiB of input, against 904.6 MiB for whole-document redaction of a 32 MiB file.

The condition is not tied to our detectors: wrapped around Presidio's recognisers, our engine made their chunked output identical to their whole-document output, and the same check exposed a Presidio pattern whose output depends on arbitrarily distant text. On the test corpora naive chunking leaked up to 237 values a whole-document pass redacts, while SafeStream matched the whole-document output under every chunking tried.

Every real-data evaluation was pre-registered, with its protocol and manifest committed before any run, and is reported whatever its direction. On 14,626 real credentials from 337 repositories (CredData), the pattern tiers fully redacted more values than gitleaks (0.55 against 0.40, robust to resampling by repository) at a higher false-alarm rate. An opt-in tier built from documented log formats redacted all 618 user names in held-out Linux logs. On court judgments, Presidio first detected more personal data than our named-entity configuration; a free-text configuration designed on development data then reached character F1 0.67 against 0.54 on 1,014 held-out judgments, and 0.76 against 0.63 on Wikipedia biographies. Intervals are paired bootstrap at the document, cluster and two-stage levels; margins that do not survive clustering, disabled scanner filters or contamination checks are reported as such. Code, protocols, manifests, raw results and per-document predictions are public.

The work concerns data protection, privacy-enhancing technologies and secure data handling, and should interest readers who build or deploy data-loss-prevention and secret-scanning pipelines.

The manuscript is not under consideration elsewhere, and all authors approve the submission. A separate manuscript by the same authors, "The Agent Holds the Key: Auditing What LLM Agents Learn Through Redaction Tools", is under review at a conference. It studies a different question, what an AI agent learns through the tool's agent interface, on different corpora, and reports no result of this manuscript; we will gladly send it to you. As stated in the manuscript, an AI assistant was used to implement evaluation scripts and to draft text; the authors reviewed every result and every sentence and take full responsibility for the content. The authors declare no competing interests.

Yours sincerely,

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behalf of all authors
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